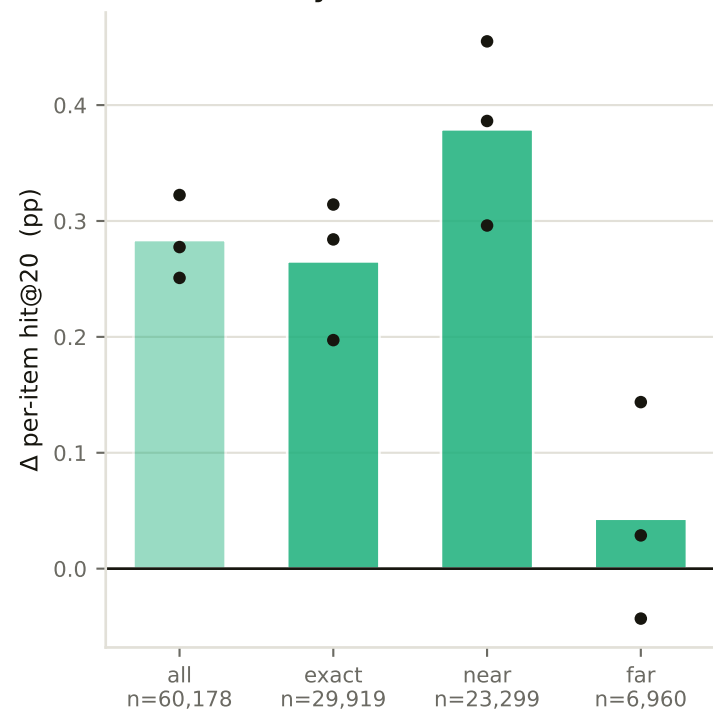


Does the object modality help **because** its vocabulary covers the catalogue?

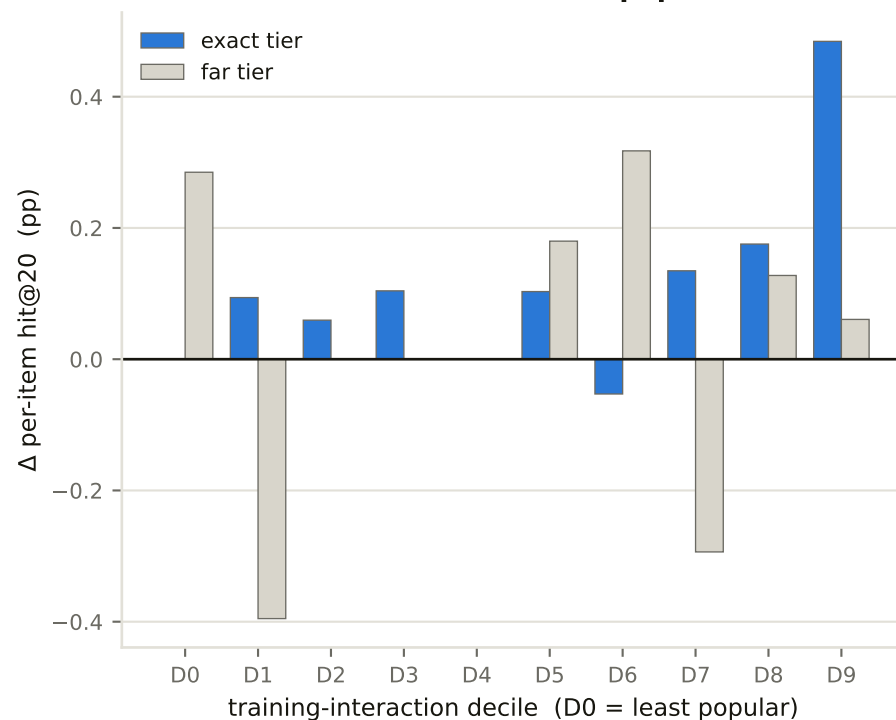
LATTICE on the shipped default `fixed` variant, seeds [0, 1, 2], paired by seed. Tiers are F17's: `exact` = the item's label is a node of the object graph, `far` = no node within cosine 0.6.

Metric is per-test-item `hit@20` (of all (user, test item) pairs in the stratum, the share where the item made that user's top-20); the offline ranking reproduces every run's logged `Recall@20` to $<3e-5$.

(a) The gain lands on the items the vocabulary reaches



(b) Same comparison inside each popularity decile, since covered items are also the popular ones



(c) Placebo: permuting which item gets which object vector

